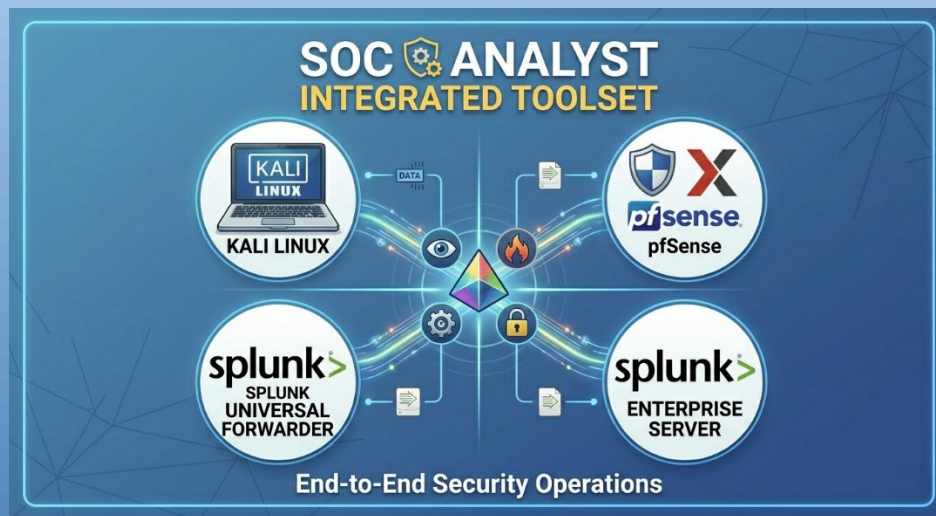




PROJECT Activity and Scope:

Design and Implementation of a Security Operations Center (SOC) to Monitoring System Using pfSense Firewall and Splunk Enterprise Tool used in project



Project Team Structure

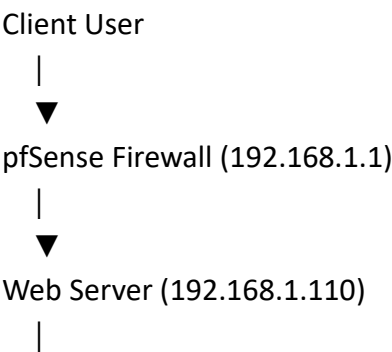
Member	Role	Main Responsibility
CHHON SOCHEAT	Team Lead / Solution Architect	Architecture design, project management, integration validation, final presentation
PENCHANTHAN VITHATTHAI	Network & Firewall Engineer	pfSense installation, VLAN/IP design, firewall rules, DHCP, NAT
PENG SEYHA	System & Web Server Engineer	Web server deployment, static IP configuration, Splunk Forwarder installation
PUM CHAMNAN	SOC Analyst & SIEM Engineer	Splunk server deployment, log ingestion, dashboard, alert creation

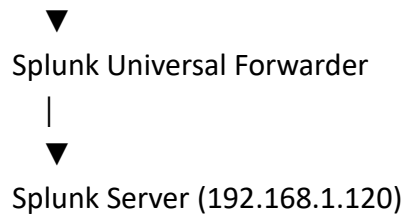
Proposed Architecture: 192.168.1.0/24

Network Components

Device	IP Address	Function
pfSense Firewall	192.168.1.1	Gateway, DHCP, Firewall Filtering
Splunk Server	192.168.1.120	SIEM, Log Collection, Monitoring
Web Server	192.168.1.110	Internal Website Hosting
Client PCs	DHCP	Access Website through pfSense

Traffic Flow





Activity Assignment List

CHHON SOCHEAT (Lead and Monitoring)

Week 1

- Define project scope
- Design network architecture
- Create project timeline
- Assign tasks to members

Week 2

- Review firewall implementation
- Review server deployment
- Conduct integration testing

Week 3

- Validate log collection
- Review dashboards and alerts
- Documentation review

Week 4

- Final testing
 - Prepare presentation
 - Demonstration rehearsal
 - Final report submission
-

PENCHANTHAN VITHATTHAI (Network & Firewall)

Week 1

- Install pfSense
- Configure WAN/LAN
- Configure DHCP

Week 2

- Configure Firewall Rules
- Configure NAT
- Restrict unauthorized traffic

Week 3

- Test web access through firewall
- Verify traffic filtering

Week 4

- Support final testing
- Prepare firewall screenshots

PENG SEYHA (System & Web Server)

Week 1

- Deploy Ubuntu/Windows Server
- Assign Static IP 192.168.1.110

Week 2

- Install Apache/Nginx/IIS
- Deploy sample website

Week 3

- Install Splunk Universal Forwarder
- Configure forwarding to 192.168.1.120

Week 4

- Generate web access logs
 - Support SOC testing
-

PUM CHAMNAN (SOC Analyst)

Week 1

- Install Splunk Enterprise
- Configure storage and indexing

Week 2

- Configure receiving port (9997)
- Verify forwarder connection

Week 3

- Create dashboard:
 - Source IP
 - Destination IP
 - Top Visitors
 - Failed Access Attempts
- Create alerts:
 - Multiple failed access
 - Suspicious IP activity

Week 4

- Create SOC monitoring report
 - Demonstrate incident investigation workflow
-

Deliverables After 4 Weeks

Network

- pfSense fully configured
- DHCP working
- Firewall policy implemented

Server

- Website hosted on 192.168.1.110
- Accessible through pfSense

SIEM

- Splunk collecting logs
- Dashboard operational
- Security alerts configured

Documentation

- Architecture Diagram
- IP Addressing Plan
- Firewall Rules List
- Splunk Configuration Guide
- Final Project Report

Recommended Firewall Rules

Source	Destination	Port	Action
LAN Users	Web Server	80	Allow
LAN Users	Web Server	443	Allow
Web Server	Splunk Server	9997	Allow
Any	Internal Servers	Unused Ports	Deny

Detail in report Implementation report